

Employee Attrition Prediction – Executive Summary

Project Title: Employee Attrition Prediction Using Machine Learning

Executive Summary

Employee turnover is one of the major challenges faced by organizations because replacing experienced employees requires significant time, cost, and effort. The objective of this project was to identify employees who are more likely to leave the company and understand the key factors influencing attrition so that Human Resources (HR) can take proactive steps to improve employee retention.

The analysis of the company's HR dataset showed that approximately **16% of employees left the organization**, indicating that while most employees remain with the company, a significant number are at risk of leaving. Among all departments, the **Sales department experienced the highest employee turnover**, suggesting that this area requires immediate attention. Within individual job roles, **Sales Representatives** had the highest exit rate, followed by **Laboratory Technicians**, making these positions the highest priority for retention initiatives.

The analysis also revealed that employees who frequently worked **overtime** were considerably more likely to leave the company. Employees with **lower monthly income** also showed a higher tendency to resign; however, salary alone was not the only reason for attrition. Other factors such as frequent business travel, job role, years of work experience, and career progression also influenced employees' decisions to leave. This indicates that improving employee satisfaction requires more than financial incentives alone.

A predictive model was developed to identify employees who may be at a higher risk of leaving in the future. The model demonstrated a good ability to distinguish between employees who are likely to stay and those who may resign, making it a useful decision-support tool for HR teams. Rather than replacing human judgment, the model should be used to help HR identify employees who may benefit from additional support or engagement.

Based on these findings, it is recommended that the company focus its retention efforts on employees working in the **Sales department**, particularly **Sales Representatives** and **Laboratory Technicians**. HR should regularly monitor employees who frequently work overtime and provide opportunities for career development, mentoring, and workload balancing. Reducing excessive overtime, improving work-life balance, and creating clear career growth opportunities can help improve employee satisfaction and reduce turnover.

Although the predictive model provides valuable insights, it has certain limitations. It cannot measure factors such as employee motivation, leadership quality, workplace culture, personal circumstances, or external job opportunities, all of which can influence an employee's decision to leave. Therefore, the model should be used alongside regular employee feedback, manager discussions, and HR experience when making retention decisions.

Conclusion

This project demonstrates how data-driven analysis can help HR teams identify employees at risk of leaving before resignations occur. By focusing on high-risk departments, improving employee engagement, reducing excessive workload, and supporting career development, the organization can reduce attrition, improve employee satisfaction, and lower recruitment and training costs. The insights generated from this project can assist HR in making informed decisions and developing more effective employee retention strategies.